

19 B Volume of each drop = $\frac{4}{3}\pi r^3$

When combined,

$$2\left(\frac{4}{3}\pi r^3\right) = \frac{4}{3}\pi r_c^3$$

$$r_c = \sqrt[3]{2}r$$

for each sphere: $V = \frac{Q}{4\varepsilon_0\pi r} = 500V$

when combined,

potential at surface $V' = \frac{2}{\sqrt[3]{2}}(500) = 790V$

